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AquaCrop-IoT: A smart irrigation platform integrating real-time images and weather forecasting

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ABSTRACT

Technological advances are providing farmers with valuable data about their crops. However, to improve resource use efficiency in agriculture, it is necessary to transform this data into practical information, applicable by farmers and/or technicians in crop management. The objective of this work was to develop a fully automated IoT platform that integrates crop images from RGB cameras with open climate data sources and crop models, to optimize irrigation strategies and enhance crop productivity under varying environmental conditions. To achieve this, the AquaCrop-IoT platform was developed, which integrates the FAO's AquaCrop model with a custom-build image capture and processing system, used to adjust the green canopy cover (CC) in real-time. Additionally, the platform incorporates weather data from in-situ weather stations, and forecasts and historical weather data from open datasets. Everything is presented in a web application that facilitates its use. The platform has been tested in a wheat crop in southern Spain throughout its growth cycle, demonstrating its potential as a decision support system for irrigation management. Dynamically updating CC values using images captured by the in-situ camera enabled the AquaCrop model to correct potential errors in crop growth estimation by including the effects of adverse factors like pests and diseases that the model cannot simulate. Furthermore, as the developed platform incorporates meteorological data daily, in real-time, it allowed the design of real-time irrigation schedules tailored to the crop in its particular environment and management. This approach improved the estimation of crop water requirements, reducing the amount of recommended irrigation water during the wheat growing season by approximately 32%.

1. Introduction

Irrigation is a key tool in increasing agricultural production, being the basic strategy to mitigate the reduction in crop yields due to the effects of drought. According to the Food and Agriculture Organization (FAO), irrigated agriculture accounts for 20 % of the global cropland and produces 40 % of the world's food supply (Alexandratos and Bruinsma, 2012). This substantial contribution underscores the importance of irrigation in ensuring food security and increasing crop yields. However, as water resources becoming increasingly scarce and stressed, it is crucial to apply irrigation precisely, tailoring water doses to the specific needs of each field to maximize irrigation water efficiency and sustainability (García et al., 2020).

Precision irrigation, based on real-time knowledge of the soil–water-plant-atmosphere system, aims to provide a solution to the aforementioned challenge. Internet of Things technologies (IoT) are essential for collecting data from sensors distributed throughout the field that can be used to generate useful information for farm management decision-making (Tzounis et al., 2017). However, relying solely on sensors is not sufficient to implement precision irrigation, and their integration with crop-water productivity models is critical to address this challenge.

Crop-water productivity models have been playing a crucial role in predicting crop yield responses to water in a wide variety of crops, serving as valuable decision-making tools to optimize irrigation scheduling and to evaluate the effects of different irrigation management strategies on crop yield (Attia et al., 2016; Douglas-Mankin et al., 2010; Raeisi et al., 2019). AquaCrop, the FAO water productivity model, is one of the most widely used (Salman et al., 2021) and it has been shown to be effective in predicting crop yield responses to water in a wide variety of arable crops (Andarzian et al., 2011; Heng et al., 2009; Linker et al., 2016; Stricevic et al., 2011).

Despite their utility, crop models like AquaCrop inherently face

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inaccuracies in estimating crop growth due to the complexity of plant-environment interactions and the effects of biotic and abiotic stressors. These models cannot account for all contributing factors. To enhance AquaCrop's performance in real-time simulations, data assimilation techniques have been explored. Such techniques integrate real-time data into models as direct inputs, for updating state variables or recalibrating parameters. For instance, Yang et al. (2023) propose a data assimilation framework to improve simulations in cases of high heterogeneity in crop phenology. However, obtaining real-time phenological data is challenging, requiring expert knowledge and costly tools. Remote sensing approaches have also been employed to automate data assimilation (Jin et al., 2020; Linker and Ioslovich, 2017; Lu et al., 2021). However, these methods are limited by the low spatial and temporal resolution of satellite images, which hinders their applicability for real-time irrigation scheduling. Similarly, Han et al. (2022) introduced an image-driven data assimilation framework that integrates convolutional neural networks (CNNs) with the ORYZA2000 crop growth model, to help smallholder farmers estimate rice yields. However, this approach primarily focuses on the model itself and does not propose an IoT solution for real-time applications; it relies on a resource-intensive CNN architecture, which poses challenges for edge-computing systems, and requires extensive calibration and labeling for each crop. To address these limitations, approaches that integrate in-situ vision sensors with the AquaCrop model could enable automatic, real-time crop growth monitoring. Such systems would facilitate dynamic model adjustments and support more effective decision-making in operational crop irrigation management.

Numerous studies in the literature highlight the application of computer vision technology in agriculture for monitoring crop growth. High-resolution color images allow for detailed analysis of plant structure and architecture. This allows high accuracy in measuring crop growth (Rasti et al., 2022). Green Canopy Cover (CC) is one of the most determining variables for simulating crop transpiration, and therefore crop growth, and can be quantified with high-resolution color images. For example, Deery et al. (2021) compared three different ground-based sensors to assess their reliability in estimating CC under varying light conditions and in the presence of dew. Other research has focused on color-based image segmentation techniques, as explored by Meyer and Neto (2008), Pan et al. (2007) and Zheng et al. (2009, 2010). Following a similar approach, Patrignani and Ochsner (2015) developed a mobile APP (Canopeo®) that uses the RGB camera of smartphones to calculate CC based on red-to-green (R/G), blue-to-green (B/G) color ratios and an excess green index (2G–R–B). Other studies have focused on water balance assessment, such as that of González-Esquiva et al. (2017) in horticultural crops using CC data calculated from low-cost cameras.

Canopy Cover is a crucial variable in AquaCrop as it determines the amount of water transpired by the crop, which in turn determines aboveground biomass production, and yield (Raes et al., 2009). Given the importance of CC in crop growth simulations, the accuracy of AquaCrop model input parameters plays a critical role in its performance. A global sensitivity analysis by Vanuytrecht et al. (2014) identified canopy development parameters as major contributors to crop yield variability, with soil parameters also playing a significant role, particularly under water-limited conditions. These findings highlight the significant impact of canopy-related parameters on model performance and underscore the value of integrating real-time CC measurements to enhance model accuracy.

Enhancing AquaCrop predictions with real-time CC data from RGB cameras, and weather variables from in-situ weather stations and open datasets, would lead to more accurate estimates of crop water needs. The objective of this work is to develop a low-cost IoT system, based on edge computing, that integrates a customized camera system to continuously monitor crop growth (CC). By leveraging data assimilation techniques, the system dynamically adjusts the AquaCrop model to correct potential past errors in simulating crop growth, thereby improving the simulation of future irrigation needs. This is achieved by integrating real-time images, real-time data from in-situ weather sensors, and both historical and forecast weather data from open datasets. These elements form the AquaCrop-IoT platform developed in this work, which was tested in a wheat crop in southern Spain.

2. Materials and methods

2.1. Location and description of the experimental plot

The AquaCrop-IoT platform was tested and evaluated on a winter wheat crop at a commercial farm located in Cordoba (Andalusia; southern Spain)(37.87 N, 4.58 W in WGS84 datum, elevation 238 m a.s.l.), a region characterized by mild winters and extremely hot, dry summers, representing typical Mediterranean climate conditions, with recurrent periods of water scarcity severely limiting irrigation water availability, and where efficient irrigation management is crucial for sustainable agricultural production. Climate data in Cordoba during the years 2001 to 2023 recorded average monthly temperatures ranging from 8.4 °C (January) to 28.1 °C (July). Mean monthly reference evapotranspiration (ET_0), calculated using the FAO-Penman-Monteith method (Allen et al., 1998), ranged from 1.1 $\frac{mm}{day}$ (December) to 7.6 $\frac{mm}{day}$ (July), with an mean annual ET_0 of 1413 $\frac{mm}{year}$, while precipitation averaged 573 $\frac{mm}{year}$ (IFAPA, 2023).

Durum wheat (Athoris variety) was sown on January 9, 2023, with a sowing dose of 200 $\frac{kgseed}{ha}$. The soil in the wheat plot was classified as Vertisol (Soil Survey Staff, 2022), it has a clay-loam texture (37.3 % clay, 34.4 % silt, and 28.3 % sand), a neutral pH (7.65; 1:2.5 H_2O) and a low electrical conductivity (0.164 $\frac{dS}{m}$; 1:5 H_2O). The maximum depth reachable by the roots was about 0.6 m.

2.2. The AquaCrop model

AquaCrop is a crop-water productivity model developed by the Land and Water Division of FAO that simulates yield response of herbaceous crops to water under varying climatic and soil conditions (Hsiao et al., 2009; Raes et al., 2009; Steduto et al., 2009). The model uses a relatively small number of explicit parameters and mostly intuitive input-variables that are either widely available or can be determined using simple methods. It takes weather data, crop and soil characteristics, and field and irrigation management practices as inputs to estimate water requirements, water productivity (WP), and crop yield.

The FAO AquaCrop model distinguishes itself from other crop models by its specific focus on water management, the use of CC instead of Leaf Area Index (LAI) as the foliage development variable, the incorporation of WP values adjusted for atmospheric evaporative demand and atmospheric carbon dioxide concentration, and the use of water stress parameters that affect a greater number of plant processes (Tenreiro et al., 2020). These features allow the model to effectively adapt and extrapolate to various locations, seasons, and even future climate scenarios. Canopy cover is used by the model to calculate separately plant transpiration and soil evaporation through the dynamic adjustment of transpiration and evaporation coefficients. From transpiration, the model estimates aboveground biomass and, subsequently, the crop yield, (Fig. 1).

The model calculates CC development using three main equations. The use of each equation depends on the stage of crop development: emergence-½ max CC; ½ max CC – max CC; onset of senescence-physiological maturity.

From the beginning of emergence until the first half of crop growth, an exponential equation (1) is applied:

$$CC = CC_0 e^{CGC \times t} \quad (1)$$

Where CC_0 is the initial CC, CGC is the Canopy Growth Coefficient and t is the time in GDD or days since sowin13g.

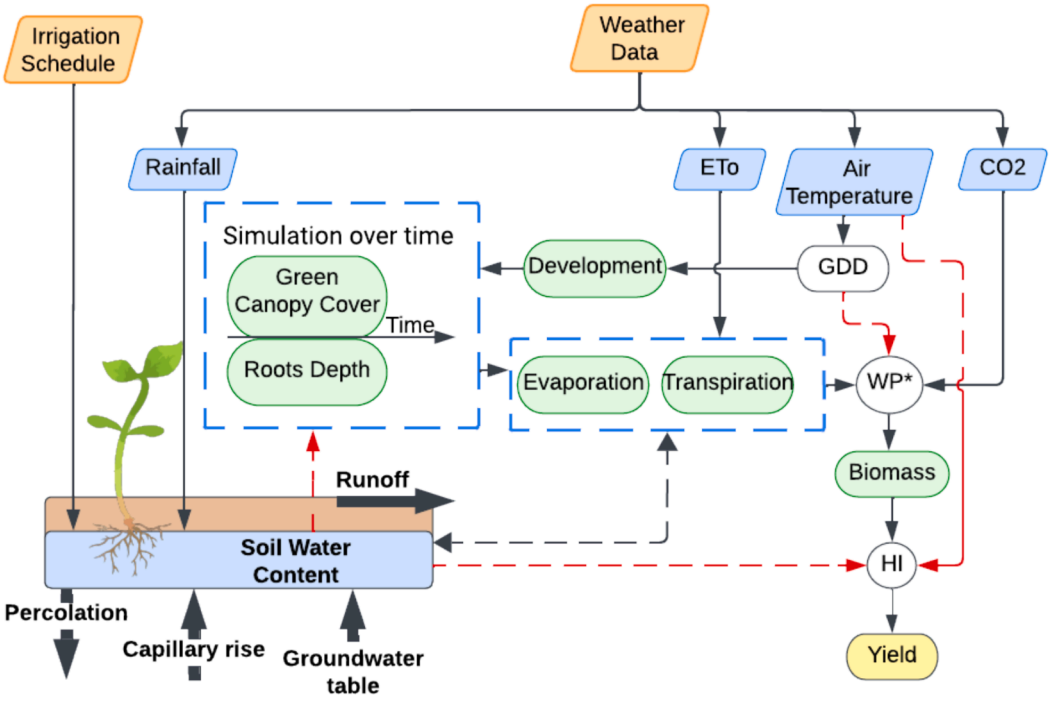


Fig. 1. Schematic diagram of the FAO AquaCrop model and the processes affected by water and heat stresses (dotted arrows). ET_0 is the reference evapotranspiration, GDD is the growing degree-days, WP* is the adjusted water productivity, and HI is the harvest index.

For the second half of crop growth, the CC decays exponentially until it reaches the maximum CC (CC_x), following the equation (2):

$$CC = CC_x - 0.25\frac{(CC_x)^2}{CC_0}e^{-CGC\times t} \quad (2)$$

Upon reaching CC_x, the canopy cover remains constant until the onset of senescence, at which point it undergoes a decline phase calculated by (3):

$$CC = CC_x\left[1 - 0.05\left(e^{\frac{3.33CDC}{CC_x+2.29}\times t} - 1\right)\right] \quad (3)$$

Where CDC is the Canopy Decline Coefficient.

These equations are for non-water stress conditions, however AquaCrop modulate this effect by adjusting the CGC (4) and CDC (5) using the following equations:

$$CGC_{adj} = Ks_{exp} \times CGC \quad (4)$$

$$CDC_{adj} = \left(1 - Ks_{sen}^{8}\right) \times CDC \quad (5)$$

Where Ks_{exp} is the water stress coefficient for expansive growth and Ks_{sen} is the water stress coefficient for the acceleration of senescence.

CGC and CDC are non-conservative parameters and depend mainly on crop variety and management (plant density).

2.3. AquaCrop-IoT: Real-time data integration into AquaCrop

To address the limitations of operating the model without real-time on-ground data, we have integrated data from diverse sources, including data from in-situ weather sensors, and historical and 7-day forecasted climate data, along with real-time information of crop growth (CC) captured by an RGB camera. The AquaCrop-IoT platform uses this data to update the CC values of the AquaCrop model, which is a modified version of the AquaCrop Python library developed by Kelly and Foster (2021), itself an extension of AquaCropOS, originally implemented in MATLAB (Foster et al., 2017), and is based on version 6 of the AquaCrop model. The library modified in this work has been tested against official AquaCrop library, to ensure identical internal behavior, and allows external control of all internal model variables, thus allowing the model to be updated with camera data.

To evaluate the use of real-time CC in simulation with AquaCrop, an updated simulation using real CC obtained from the camera images was compared with a not updated simulation running AquaCrop in its traditional way. In both simulations, the model was run from sowing to seed maturity.

2.3.1. Real-time estimation of green canopy cover

Crop growth was monitored with a 16 MP Arducam camera equipped with a Sony IMX519 sensor (Fig. 2), which was installed in the wheat field. Throughout the crop cycle, each day the camera took a photo at solar noon, when the sun's rays are closest to perpendicular to the ground, to eliminate shadows-induced errors. The images were then sent to a single-board computer (SBC)(Raspberry Pi 4 with 8 GB RAM) that calculates the CC using an image-based segmentation method, which we have developed using the C++ programing language.

This segmentation method is inspired by the approach used by Canopeo (Patrignani and Ochsner, 2015), which employs a selection of pixels based on R/G and B/G ratios (Liang et al., 2012; Paruelo et al., 2000) as well as the excess green index (ExG)(Chen et al., 2010; Richardson et al., 2007). To ensure accurate CC segmentation, these indices have been visually validated and compared with other indices, such as the normalized difference vegetation index (NDVI) (Pérez et al., 2000) and the Normalized Green–Red Difference Index (NGRDI)(Hunt et al., 2005). Among them, the combination of R/G, B/G and ExG has shown better performance for our specific use case and camera.

The segmentation library iterates over all pixels in the image, classifying each of them as either plant or soil based on the conditions outlined in equation (6). At the end of this process, the output is a binary image containing only two values: 0 for soil and 1 for vegetation. The CC is then calculated as the quotient between the pixels classified as vegetation and the total number of pixels in the image. A pixel (R, G, B) is classified as vegetation if it satisfies the following conditions:

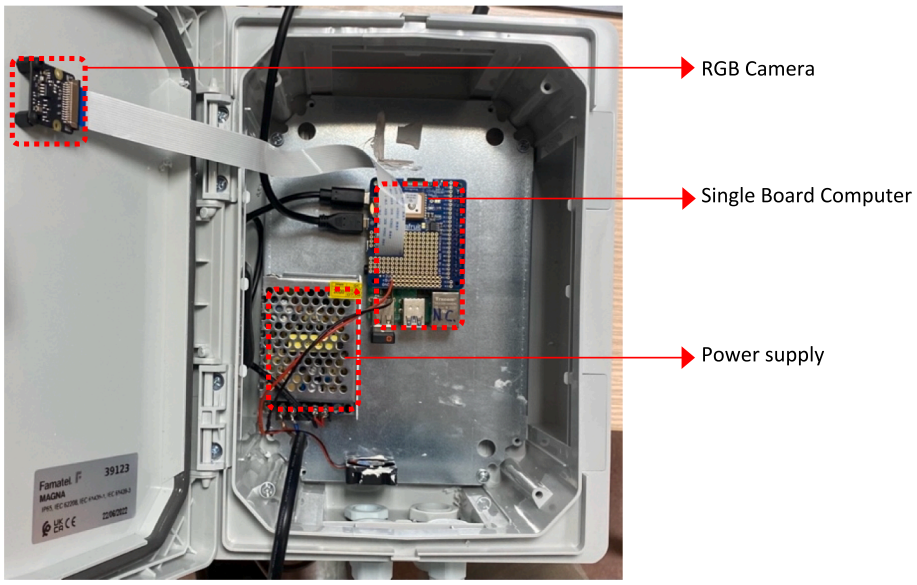


Fig. 2. Camera system hardware: a waterproof enclosure containing the single-board computer, camera, and power supply.

$$\text{Vegetation If } \frac{R}{G} < P_1 \text{ and } \frac{B}{G} < P_2 \text{and} 2\text{G} - \text{R} - \text{B} > P_3 \quad (6)$$

Where P_1 and P_2 are parameters typically close to 1, and P_3 is typically close to 20. These settings depend on the type of camera and the visual conditions and should be manually tested by visually comparing the segmented image with the real one to determine the optimal settings.

2.3.2. Climate data acquisition and forecasting

The system uses real-time data from an ATMOS 41 weather station (Fig. 5, c) installed in the field. This weather station has twelve sensors that measure solar radiation, precipitation, air temperature, barometric pressure, air vapor pressure, air relative humidity, wind speed and wind direction, and transmits all the data to an IoT platform (Puig et al., 2022) in the cloud every 5 min.

Historical and forecast weather data are collected from the Spanish State Meteorological Agency (AEMET) and integrated into the platform through its API (Application Programming Interface). AEMET offers weather forecasts for the next seven days and long historical data series for more than 30 years.

2.3.3. Data assimilation and simulation processes

Data assimilation and simulation processes covers three stages (Fig. 3) according to the different datasets used: i) Stage 1: simulates from sowing to the current date, using the meteorological variables measured at the in-situ weather station as inputs to the model, and updating the CC using the average values measured by the cameras; ii) Stage 2: simulates from the current date to the next seven days, using the forecast weather data provided by AEMET, and iii) Stage 3: simulates from the seventh day from the current date to harvest, using the average of the historical weather data provided by AEMET for the last 20 years.

Starting from the sowing date, in Stage 1 AquaCrop is run using real-time data from the RGB cameras and the in-situ weather station. This means that after each simulation step, the CC estimated by AquaCrop is replaced with the actual CC value calculated from the camera images. Starting with today's simulation, in Stage 2 the model uses the weather forecast for the next seven days to simulate crop growth. During the remainder of the simulation period, the model uses historical weather

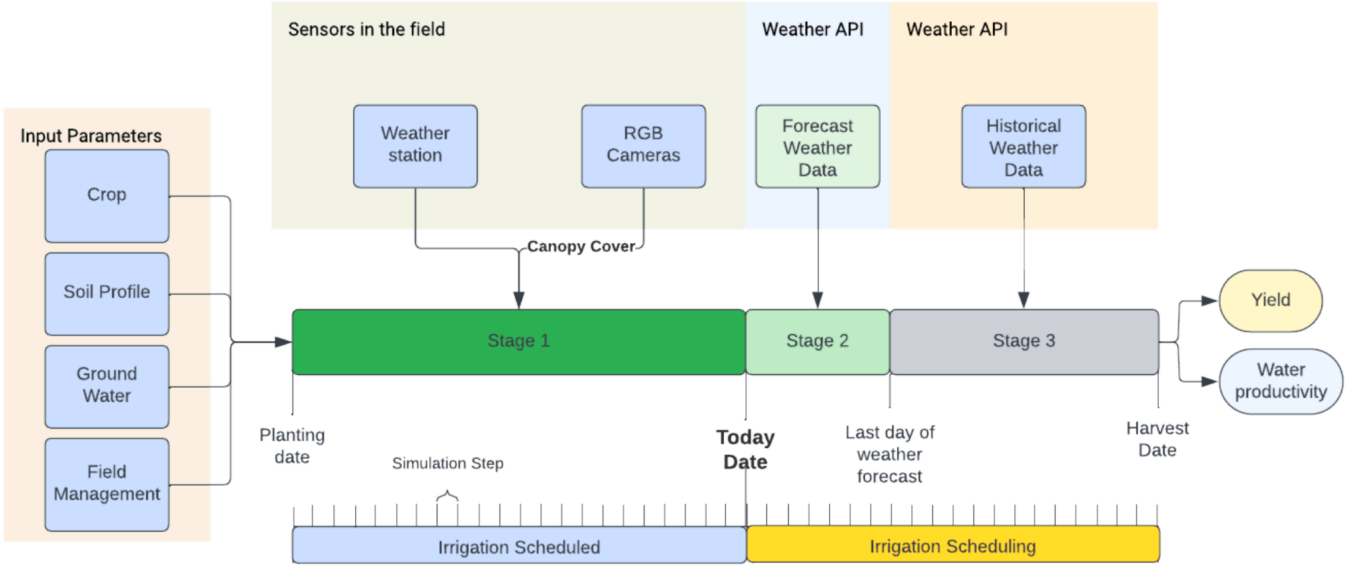


Fig. 3. Diagram of the AquaCrop-IoT system data assimilation and simulation processes.

data to estimate CC and accurately simulate the water balance and estimate irrigation needs.

Different irrigation management strategies can be used. To define these strategies, users can choose among various criteria and approaches included in AquaCrop's irrigation module: (1) Keep soil water content at a Soil Moisture Threshold (percentage of Total Available Water; TAW); (2) Establish time (fixed time interval or soil moisture threshold); (3) Irrigation depth criteria (fixed irrigation depth or back to field capacity), and (4) Predefined Schedule (daily irrigation depth is supplied by the user). In Stage 1, since the final decision on when and how much to irrigate is always made by the users, it is essential that they facilitate past irrigation events. This ensures that the model can accurately simulate the water balance and generate reliable recommendations. For Stages 2 and 3, the module calculates irrigation needs daily based on the strategy chosen by the user.

All simulation results, including water needs, recommended irrigation, crop growth, biomass, and estimated yield, are saved in a database for quick access by the user, who can then decide whether to continue with the current strategy or change the irrigation schedule and run the simulation again to find a better one.

2.3.4. System architecture

To achieve the proposed objectives, the system architecture must be adaptable to varying environmental conditions, require minimal maintenance and incorporate cost-effective components. The architecture is divided into the following four layers (Fig. 4): i) Perception layer includes one or more cameras that collect information from the crop; each camera is controlled by an SBC that manages and processes the data; ii) Transport layer connects the cameras to the main system using an Ethernet cable; iii) Processing layer is in charge of controlling all the cameras, performing the simulations and storing the information, and iv) Application layer is a web interface that users can access on their devices to view real-time information about their crops.

This architecture is based on edge computing, meaning that the perception, transport, and processing layers are located in the field. By processing the data on-site using SBCs, the cost of data transmission is significantly reduced. A Raspberry Pi 4 SBC was used in this study. However, any SBC with similar characteristics that supports the Linux operating system could also be used. This is especially important in agriculture, where sensors are often installed in remote locations with poor connectivity. In these situations, it may be possible to use communication systems such as LPWAN to send only the processed data to the cloud. However, in this study, we opted for a 4G communication system to transmit images to the web page, allowing us to visually validate the accuracy of CC segmentation. Additionally, in cases where there is no connectivity, it is still possible to connect directly to the IoT system from the farm.

Cameras are the main component of the system (Fig. 5, a). They take pictures of the crop, which are used to calculate in real-time its growth. The system can scale to include one or multiple cameras, each of which operates independently. Each camera system is installed pointing to the ground. Once a day, the camera takes a photo and sends it to a Node.js server (NodeJS [WWW Document], 2023) hosted on an SBC. The server uses the canopy cover module to calculate the CC from the image and saves the information in a MongoDB database. The photo is also saved in a NFS (Network File System) folder that is shared with the main server installed in the processing layer. This allows easy downloading of all images from the main server, which is located in a more accessible area compared to the camera system, making image retrieval more convenient for research purposes. The processing layer consists of another SBC (Fig. 5, b) and a switch that allows connecting multiple camera systems. On this SBC, a modified version of the AquaCrop model is deployed and is ready to be accessed by the main server. The server managing this layer is developed using Node.js and incorporates a 4G modem for cloud accessibility. However, any technology capable of establishing an Internet connection with an SBC can be used. This allows users to interact with the system remotely. To connect with IoT sensors, the server is integrated with an IoT platform developed by Puig et al. (2022). This platform is built on the LD version of FIWARE (Cantera et al., 2019), which offers enhanced API standardization. As a result, this architecture is inherently compatible with other IoT platforms adhering to the same standards. The IoT platform efficiently manages all aspects of sensor data, including collection, cleansing, governance, and fusion, ensuring reliable and consistent data processing.

The cost of the hardware components of the IoT platform, including essential accessories like cables and waterproof cases, is relatively low, but it can vary depending on factors such as stock availability. In this study, the current total cost per camera system does not exceed €150, while that of the main server is below €200.

Finally, a web application using the ReactJS library (ReactJS [WWW

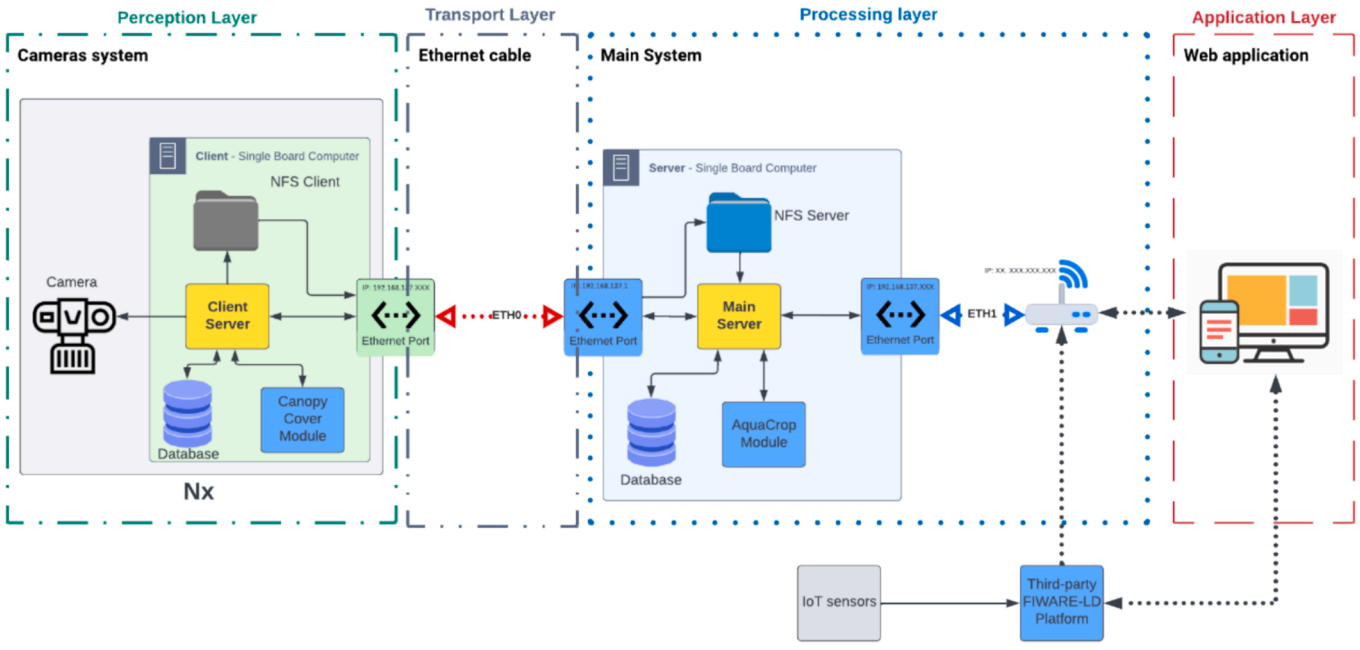


Fig. 4. Diagram of the AquaCrop-IoT Cameras System Architecture.

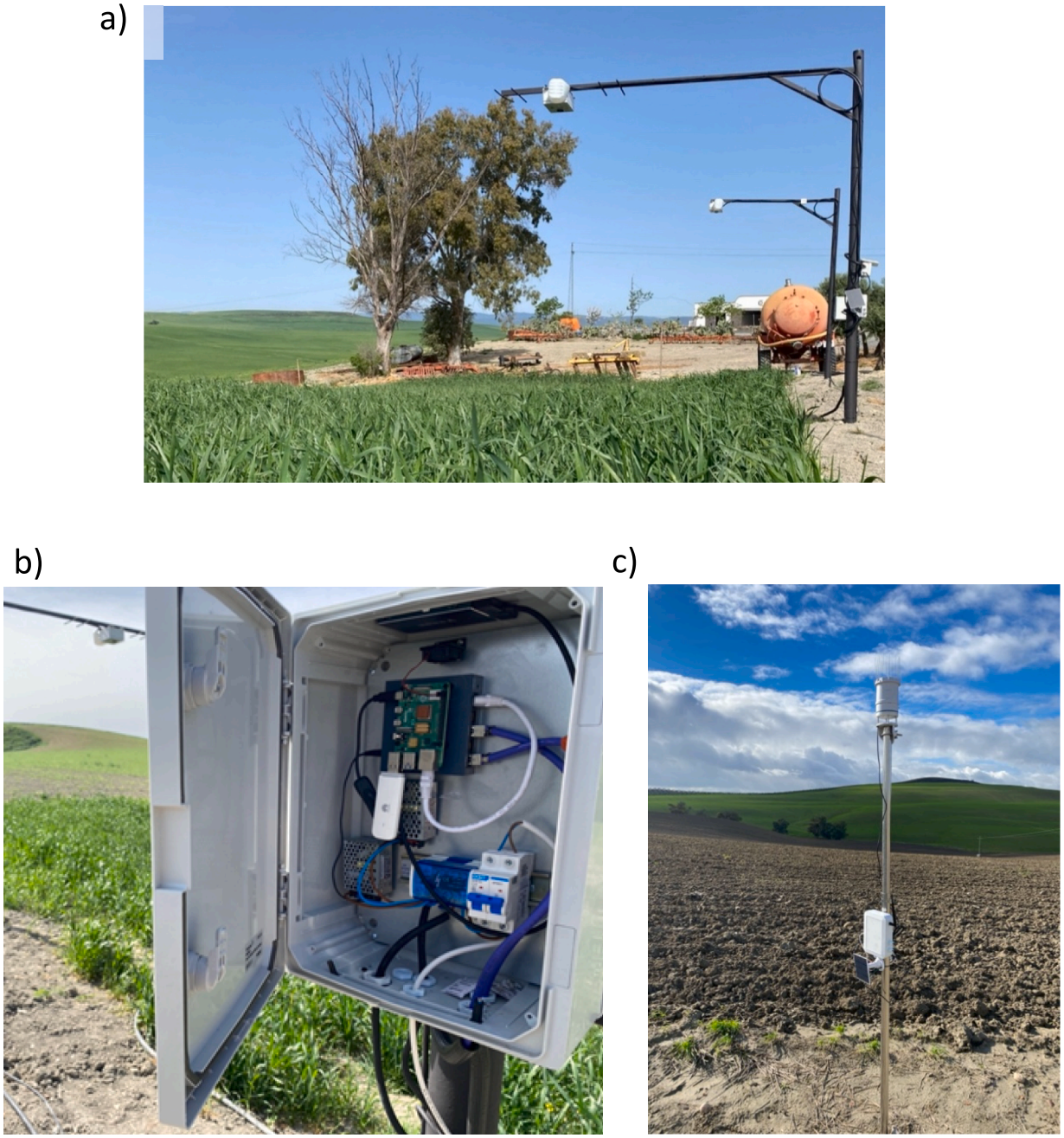


Fig. 5. IoT system hardware components: a) The two cameras mounted in the field. b) Waterproof box installed on one of the posts, housing the main server. c) All-in-one weather station installed in the field.

Document], 2023) was developed to provide users with a friendly interface. In particular, the platform includes two main components: a visualization dashboard that displays the information collected by the sensors, and a control panel that allows users to view, interact, and analyze crop images and crop growth simulations. When the user first starts using the system, all input parameters required by the model (soil and crop parameters, irrigation schedule and crop management) must be provided. These parameters are saved in a MongoDB database hosted on the same SBC.

2.3.5. System implementation

A camera was installed on the wheat plot, mounted 3 m above the ground on a metal pole, (Fig. 5, a). This allowed photos to be taken within a FOV (Field of View) of 3.4 × 4.5 m with a resolution of 4656 × 3496 pixels. In addition, a weather (Fig. 5, c) was installed at a height of 1.8 m. The in-situ weather station was connected to a datalogger that sent the data to the cloud via a LoRa network. This data was stored in a cloud database and used in simulations by the AquaCrop model.7.

3. Results and discussion

3.1. Performance of real-time green canopy cover data assimilation

Fig. 6 shows precipitation and ET_0 data from the in-situ weather station along with the soil water depletion in the root zone (D_r) simulated by AquaCrop. The simulation includes two scenarios: one adjusted using the CC calculated by the cameras (CC updated) and one without this adjustment (CC not-updated). The data reveals an exceptionally dry year but marked by heavy rains in December that caused a delay in sowing until January. Between sowing date and May, cumulative

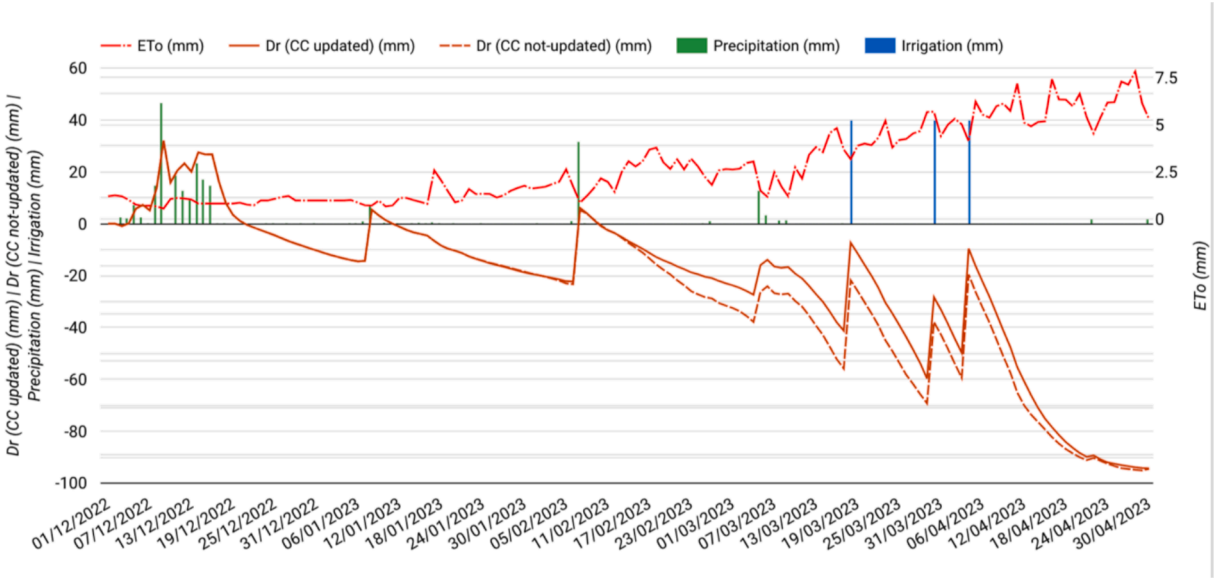


Fig. 6. Applied irrigation and precipitation (mm day^{-1}), daily ET_0 (FAO Penman-Monteith method) from the in-situ meteorological station, and soil water depletion in the root zone (D_r), represented as: $\sum[\theta_V(i) - FC(i)] \times LT(i)$ where $\theta_V(i)$ is the volumetric soil water content at each measurement date (mm), FC is the soil field capacity (mm), and LT (mm) is the thickness of the soil layer at depth interval i.

rainfall barely reached 60 mm, well below the average for this area of 237 mm for the same period during 2001–2023, while accumulated ET_0 reached 381 mm, well above the average of 279 mm for the area during 2001–2023. Despite the crop's high-water requirements, only three irrigation events of 40 mm each were applied, on March 18, March 30, and April 4, 2023, due to water scarcity during the study season caused by the situation of severe drought in the area. As a result, D_r increased (indicating greater soil water depletion), leading to severe water stress towards the end of the season. Additionally, the simulation with CC updated shows a lower D_r starting on February 13 as compared to the simulation with CC not-updated. This difference was due to an earlier crop emergence in Aquacrop simulation (CC not-updated) compared to what was observed in the field, resulting in an earlier and higher CC, and greater crop growth and greater soil water extraction (higher transpiration) than in the simulation with CC updated (as can be seen in Figs. 8 and 9).

Fig. 7 shows photos of the crop canopy taken on three different dates, March 11, April 3, and April 30, 2023. The estimated CC was 54 %, 90 %, and 24 %, respectively. The optimal settings for the RGB camera were 1 for the R/G and B/G parameters, and 15 for the green excess index parameter. These settings were selected through a manual comparison between the segmented images and the original RGB images. While this method ensures accurate segmentation under constant conditions, it may require adjustments if there are significant changes in visual conditions. To enhance adaptability to such situations, future work should explore the integration of machine learning-based methods, such as the one proposed by Puig et al. (2025), for automatic adjustment to varying environmental conditions, thus offering a more dynamic solution without the need for manual recalibration.

Fig. 8 shows the daily evolution of CC using the CC values observed from camera images in real time (CC updated, hereafter) and CC values simulated by AquaCrop for the winter wheat crop (CC not-updated, hereafter). It also shows the evolution of the simulated aboveground biomass under both scenarios. AquaCrop predicted crop emergence on January 21, however it was not until February 8 that emergence was observed in the field. It is also observed that the onset of crop senescence was three days earlier in the simulation with CC-not updated, but with a less pronounced decline in CC during the final phase of crop, controlled by CDC, a non-conservative parameter (section 2.2).

Since the AquaCrop model (CC not-updated) predicted earlier wheat

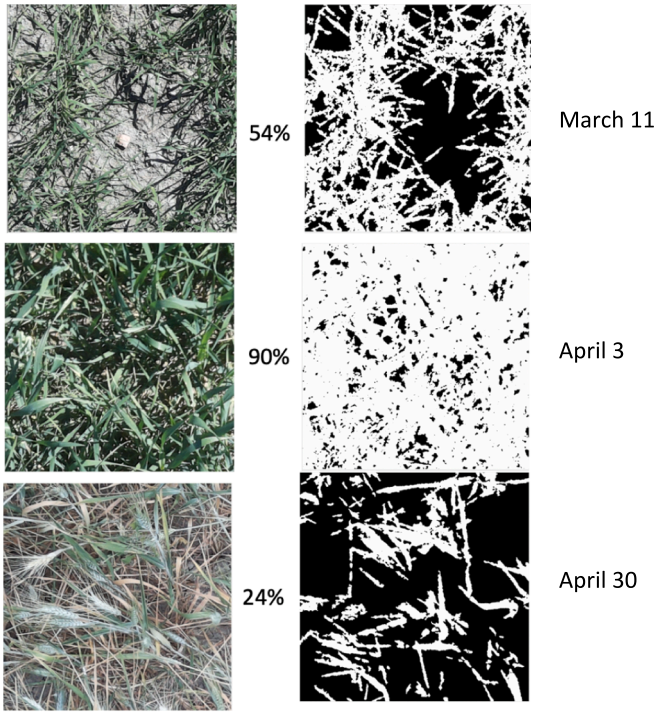


Fig. 7. Canopy cover on different days of the wheat crop cycle: Images on the right are those converted from the original photos, with white pixels representing the green color of the vegetation.

emergence than observed in the field and consequently earlier and higher CC during the crop cycle, particularly in the early development stages, compared to in situ measurements (CC updated), the simulated crop transpiration with CC not-updated was higher and soil evaporation lower than in the simulation with CC updated (Fig. 9). Until the onset of crop senescence, accumulated ET was 400 mm (147 mm transpiration and 253 mm soil evaporation) in the CC updated simulation, while in the CC not-updated simulation it was 412 mm (180 mm transpiration and 232 mm soil evaporation). The higher crop transpiration simulated by

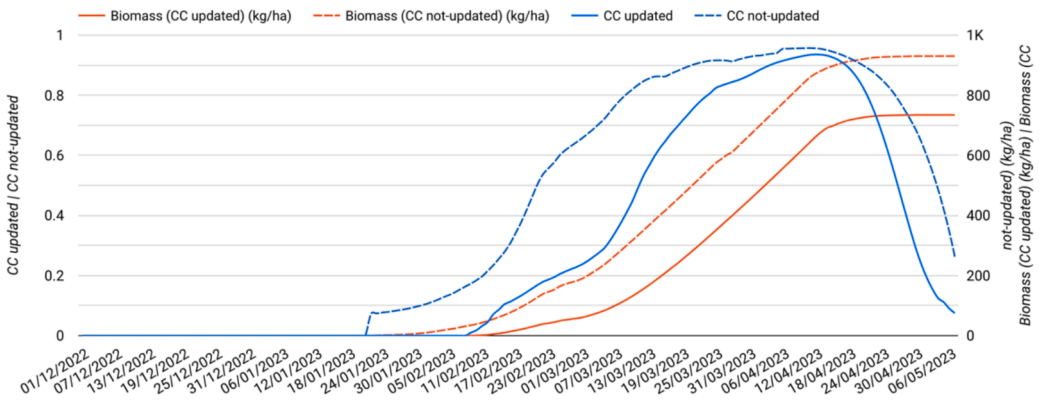


Fig. 8. Evolution of simulated green canopy cover and biomass, updated with observed CC values, versus not- updated CC. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

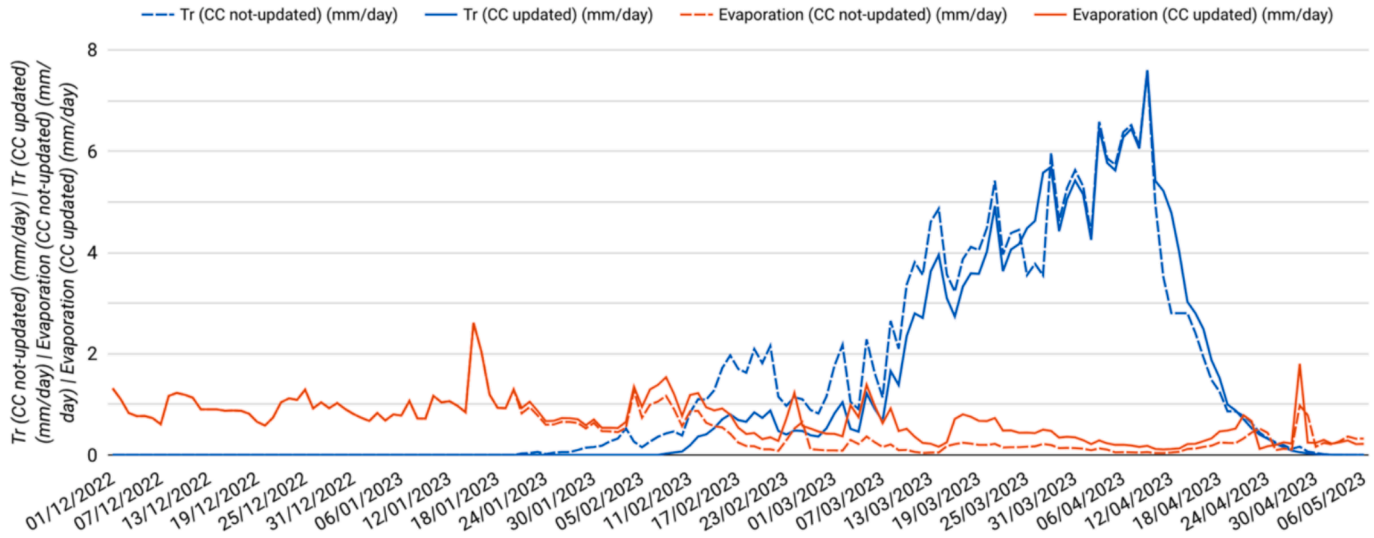


Fig. 9. Evolution of simulated crop transpiration (Tr) and soil evaporation, in the two simulations with updated versus not-updated CC.

the model with CC not-updated led to a higher estimate of aboveground crop biomass (Fig. 8), which AquaCrop uses to calculate crop yield by multiplying it by the corresponding harvest index.

The earlier crop emergence simulated by AquaCrop (CC not-updated) and the resulting increases in simulated CC, transpiration and biomass may be due to a variety of factors. These include difficulties in accurately defining model input parameters (e.g. soil properties), the misalignment of calibrated crop parameters, which although selected to be representative, have been calibrated at other locations and with other crop varieties, or simplifications employed by the model to reproduce the complex interactions between plants and their environment (e.g. simplifying physiological processes into mathematical equations, and not considering some possible abiotic or biotic stresses). Thus, adjusting CC in real time (CC updated) allows for more accurate predictions, by correcting for past errors in crop growth simulation. This has been demonstrated in several works using data assimilation techniques with remote sensing technology (Jin et al., 2020; Lu et al., 2022; Mohamed Sallah et al., 2019). These approaches have shown enhancements in model predictions and adaptations, with less reliance on extensive model calibration processes. However, they are negatively affected by the low spatial and temporal resolution of satellites, which can be aggravated by adverse meteorological conditions such as cloud cover.

It is important to note that in our procedure, although the AquaCrop model was adjusted in real-time using the measured CC data, it was not calibrated, meaning that its internal behaviour remained unchanged. In future work, deep learning techniques could provide innovative solutions to reduce uncertainties and improve model accuracy. For example, integrating deep learning methods to monitor factors such as the presence of weeds, pests, or other stressors in real time could enhance the model's adaptability without compromising its internal functionality.

3.2. *AquaCrop-IoT system implementation for irrigation scheduling*

The AquaCrop-IoT platform's greatest utility lies in its potential to be used for operational decision-making. It improves the performance of the model using data provided by sensors. This opens possibilities to generate and analyze real-time irrigation programs, aimed at maximizing crop yield and irrigation water productivity. To illustrate, consider a hypothetical scenario where an agricultural expert consults the AquaCrop-IoT system on March 16, 2023, to obtain irrigation recommendations for the next seven days. To highlight the advantages of using the AquaCrop-IoT simulation, a parallel AquaCrop simulation is conducted without updating the CC values. The expert selects a "Moisture Stage Target" irrigation strategy, aiming to maintain the soil water depletion (SWD) at 90 % of the total available water (TAW) and setting a maximum irrigation application of 20 mm per day.

Table 1 shows the irrigation dates and doses calculated in the two crop growth simulations performed. The AquaCrop-IoT simulation (CC updated) recommended three irrigations, with a total depth of 44.1 mm, whereas the simulation with CC not-updated values recommended four

Table 1
Irrigation recommendations from the AquaCrop-IoT system versus a simulation with not-updated CC values, using March 16, 2023, as the reference date.

Date	Irrigation recommendations (mm)	
	AquaCrop-IoT simulation	Not-updated CC simulation
16/03/23	25.0	25.0
17/03/23	0	11.5
18/03/23	0	0
19/03/23	0	7.7
20/03/23	9.6	0
21/03/23	0	8.8
22/03/23	9.6	0
23/03/23	0	11.8

irrigations, with a total depth of 64.8 mm. As explained in Section 3.1, the earlier crop emergence simulated by the model, relative to that observed in the field, resulted in an overestimated CC. This results in erroneous calculations of both transpiration and soil evaporation. Specifically, the simulation with CC not-updated estimated transpiration at 64.8 mm, while the simulation with CC updated (with the actual, delayed emergence date) estimated it at 34.2 mm. In contrast, the simulation with CC not-updated underestimated soil evaporation, calculating 24.1 mm compared to 39.0 mm in the simulation with AquaCrop-IoT. These differences are due to the fact that in the simulation with CC not-updated the model simulated that crop growth started earlier than it did, thus covering the soil more quickly and therefore modifying the distribution between soil evaporation and crop transpiration in the early stages of crop growth, which is when the few rainfall events occurred.

To evaluate soil water balance and improve irrigation scheduling, the use of continuous soil moisture sensors would help to adjust the water balance, allowing crop growth to be adjusted. However, these sensors are expensive, have a low volume of influence, thus requiring a considerable number to adequately represent soil water content, and are complicated to install in annual crops, as they must be removed for various agricultural practices (e.g. tillage). In addition, they are not suitable for installation in certain soil types (e.g. with expansive clays).

In this study, two moisture sensors, METER Group Teros 12, were installed in the soil and integrated using the same data logger used for the weather station. However, due to the lack of rainfall during the crop season and the expansive nature of the soil (Vertisol) in the area where the crop field was located, the sensors recorded erratic and unreliable values. This problem was due to the formation of air cavities around the sensors, caused by the shrinkage and cracking of the soil as it dries. These cavities disrupted the contact between the sensor and the soil, compromising the accuracy of the moisture measurements, which is why the data from the soil moisture sensors were not used as input for the model.

3.3. Web application service

A web application has been developed to make the AquaCrop-IoT platform more user-friendly. This allows users to run it in an easy and intuitive way. To achieve this goal, the interface is divided into different pages, each one focused on a particular aspect. The “Devices” page manages and controls IoT sensors, like the weather station. Users can easily add, delete, update devices, display and download information (Fig. 10). The “Advanced Search” page serves as a visualization tool, enabling users to compare values across different devices. In this case, only one weather station has been installed.

There is a specific page for the information collected from the cameras (“Camera” page), which serves as the central hub for camera management and its interactions with the AquaCrop model.

The system is equipped with four submenus that facilitate viewing and control: i) The “Information” submenu displays essential system metrics such as storage utilization percentage, CPU temperature, and RAM usage; this information enables users to verify that the system is functioning as expected (Fig. 11, a); ii) The “Images” submenu (Fig. 11, b) showcases all recorded images and estimated CC values; users can view images captured by the cameras (Fig. 11, c) and the calculated CC derived from the segmentation algorithm; iii) The “Crop simulation” submenu (Fig. 12) provides comprehensive information on crop growth and irrigation strategies generated using AquaCrop, and iv) The “Settings” submenu allows users to specify input model parameters (Fig. 13).

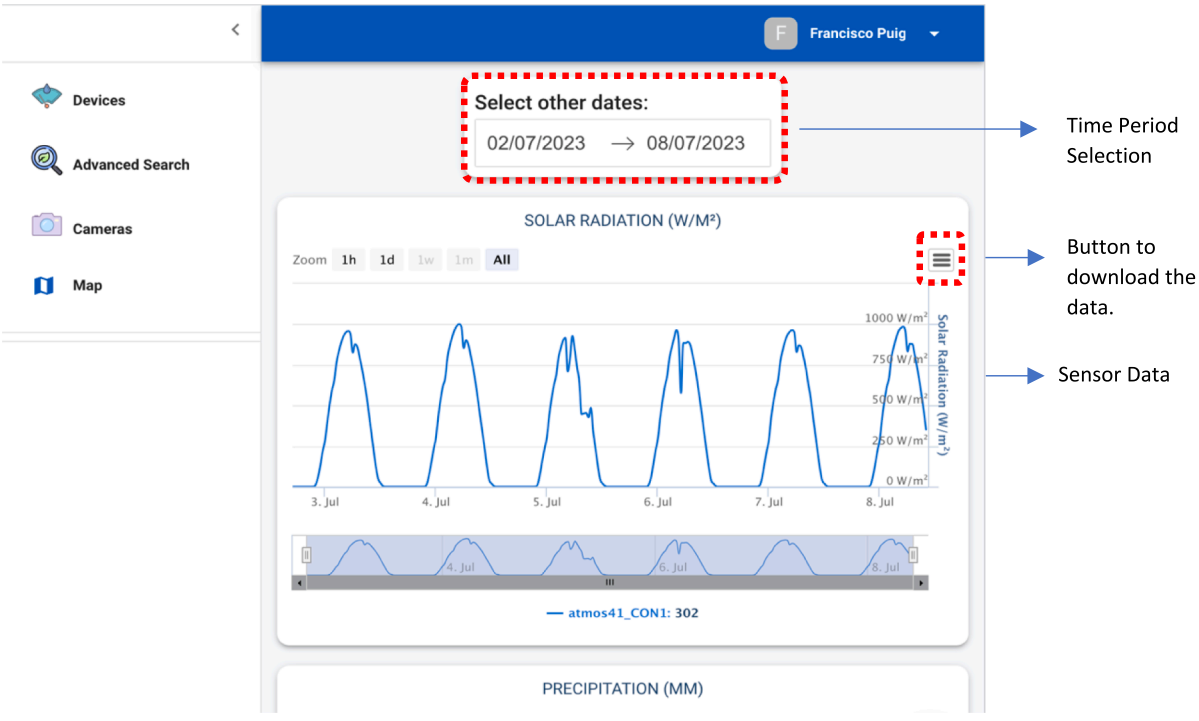


Fig. 10. Devices Page of the Web Application.

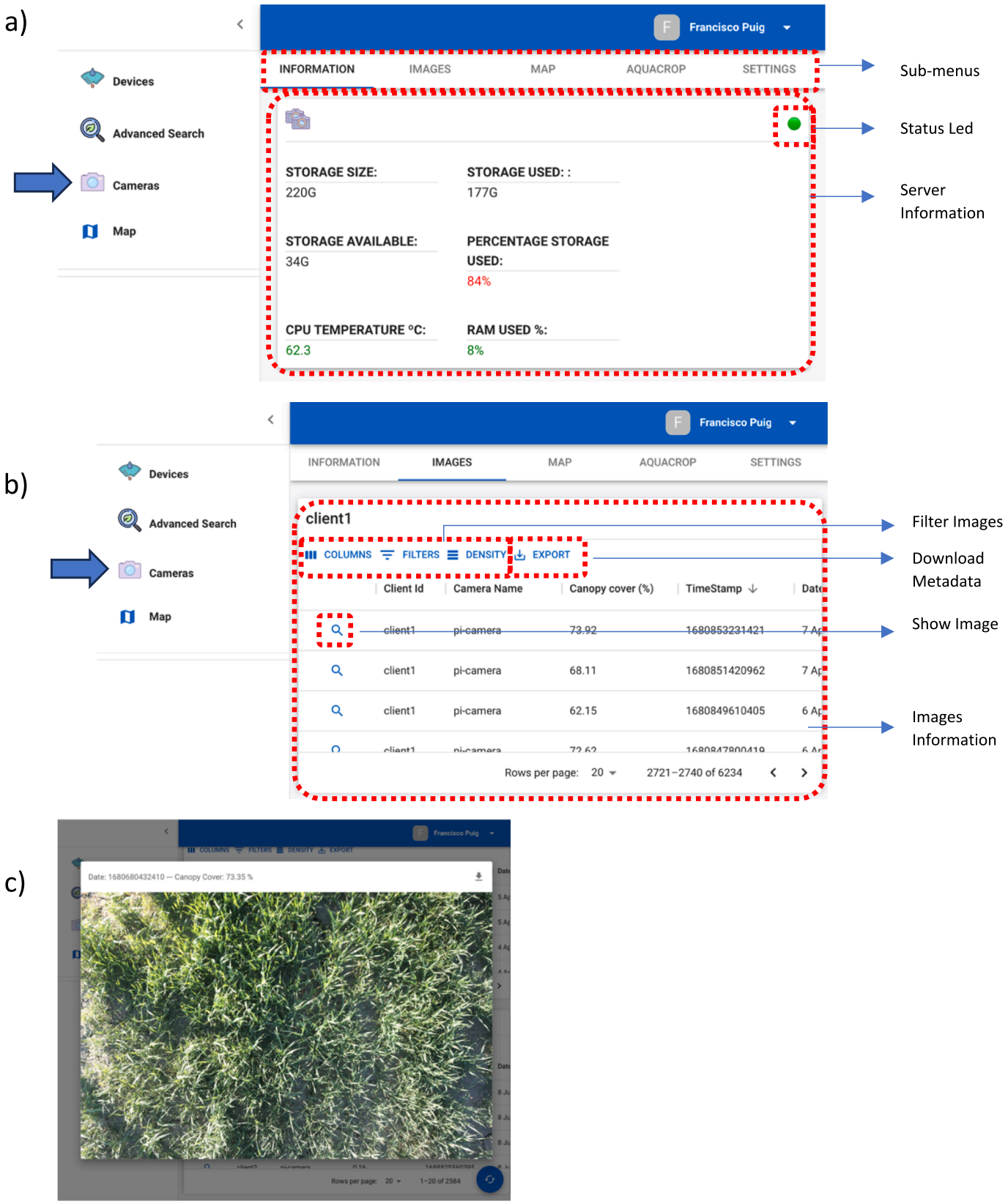


Fig. 11. Information and images pages of the Web Application: a) Information Section: displays details on the camera page, including key specifications and metadata; b) Images Section: highlights the gallery of images associated with the camera, and c) Image View: provides an enlarged, detailed view of a selected image from the gallery in the Images section.

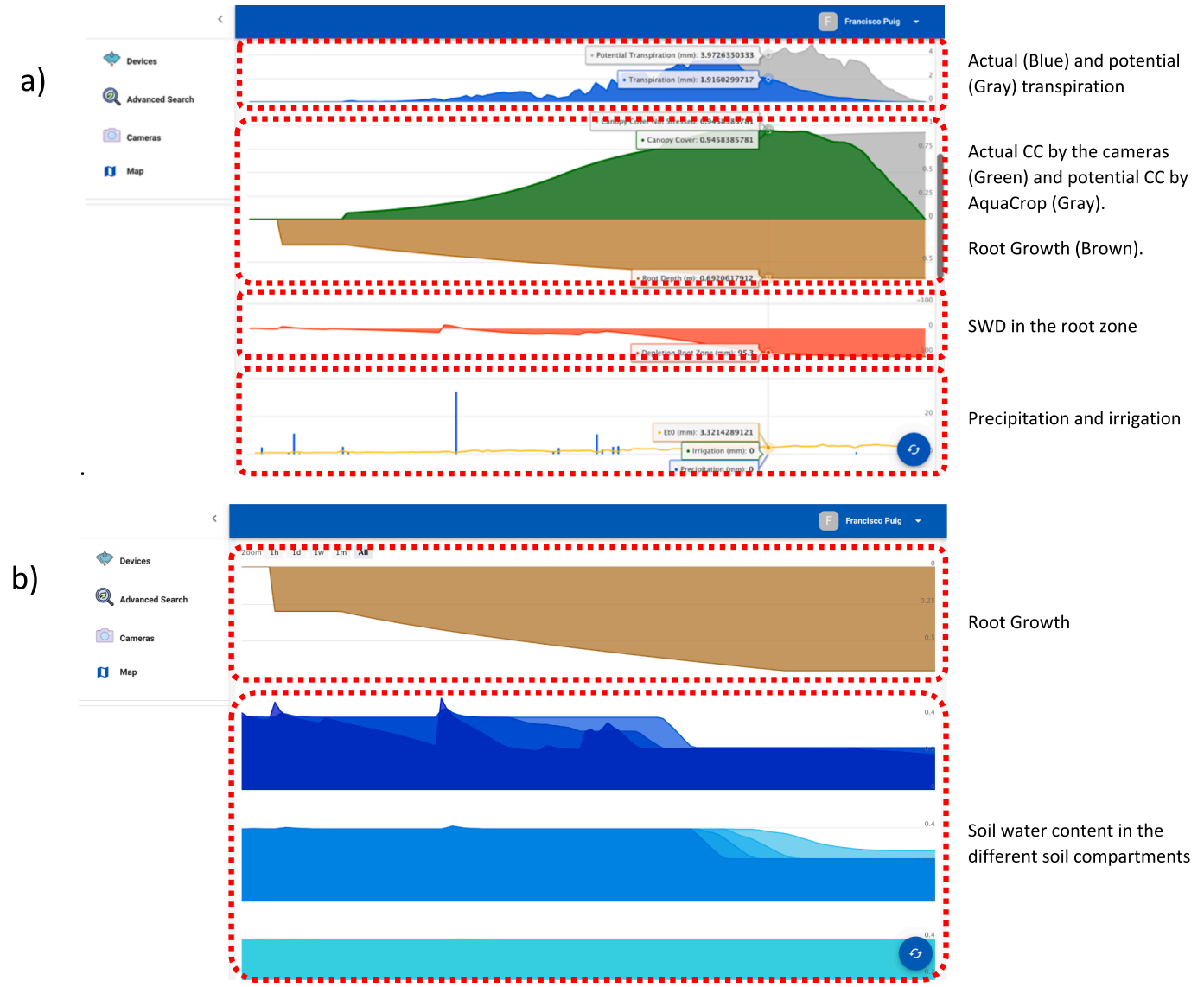


Fig. 12. AquaCrop simulation results page of the Web Application: a) Graphs showing AquaCrop simulation results: crop transpiration (blue line for actual, grey line for potential), CC growth (green graph), root growth (brown graph), soil moisture depletion (red graph), and rainfall trends, and b) Graphs depicting AquaCrop-simulated root growth and soil water content in various soil compartments.

The information in the "Crop simulation" submenu is presented in the form of two tables and two graphs. The first two tables show general information and irrigation recommendations for the next seven days. The first graph (Fig. 12, a) provides a complete visualization of key variables such as CC, root zone SWD, ET_0, irrigation applied and rainfall. This graphical representation helps monitor crop growth and irrigation requirements. The second graph (Fig. 12, b) shows soil water content for each soil compartment and root growth.

The last sub-module (Fig. 13) is designed to configure both cameras and model settings, enabling users to conveniently adjust parameters to suit their specific requirements. For example, users can modify irrigation strategies and immediately view the updated results within the AquaCrop sub-menu.

4. Conclusions

Digitization and sensors for irrigation management are useful tools that provide relevant information for better use of agricultural resources. However, this information requires processing and interpretation before it can be directly applied. Crop simulation models play a relevant role in translating this data into formats useful for end-users in decision-making, but they are not well adapted for real-time scenarios. In this work, an edge-computing IoT platform has been developed, which integrates a custom camera system capable of monitoring crop growth in real time. This data is then used to update AquaCrop (the FAO crop model to simulate yield response to water) by adjusting CC on a daily basis. Furthermore, the system integrates weather data from an in-field meteorological station and complements it with historical and forecast data from open-source weather datasets. The Aquacrop-IoT platform has been implemented in a user-friendly application. The system was tested on a wheat crop. By updating the CC using actual data measured in the field, the system dampened past errors in AquaCrop simulations, thus improving the calculation of crop transpiration and irrigation requirements. As a result, the estimation of the volume of irrigation to be applied to the crop was reduced by approximately 32 %. This Aquacrop-IoT platform represents an improvement in real-time irrigation scheduling, something little explored to date.

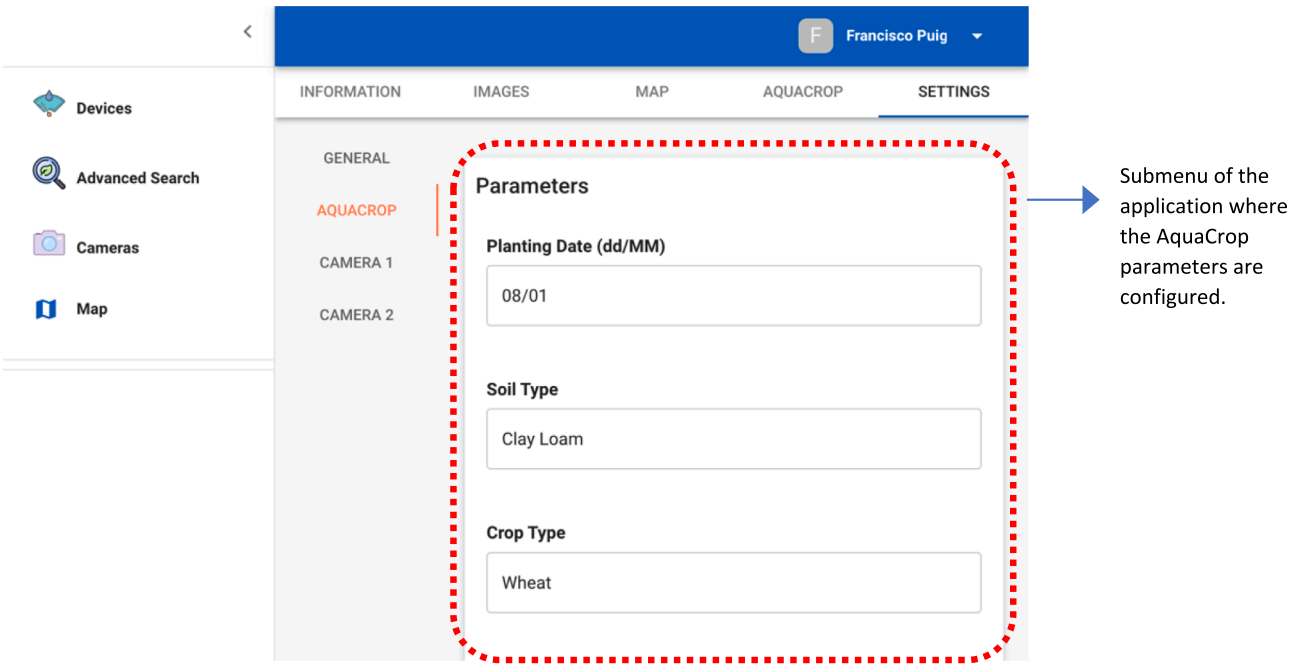


Fig. 13. AquaCrop model settings page of the Web Application.

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CRediT authorship contribution statement

F. Puig: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **M. Garcia-Vila:** Writing – review & editing, Investigation, Conceptualization. **M.A. Soriano:** Writing – review & editing, Investigation, Conceptualization. **J.A. Rodríguez Díaz:** Writing – review & editing, Supervision, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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